Data Exploration

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Notes

Please refer to the codebook for information on the data, especially on understanding what the variables mean.

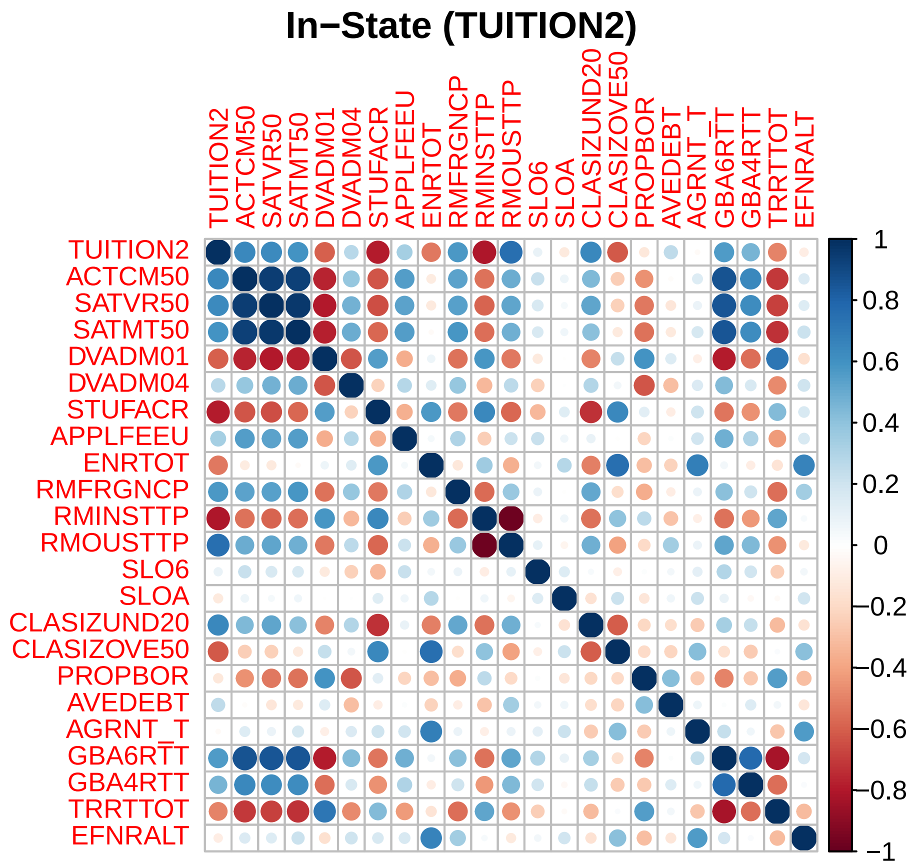
Predict TUITION2 and TUITION3 (In-state/Out-of-state Tuition)

Helper

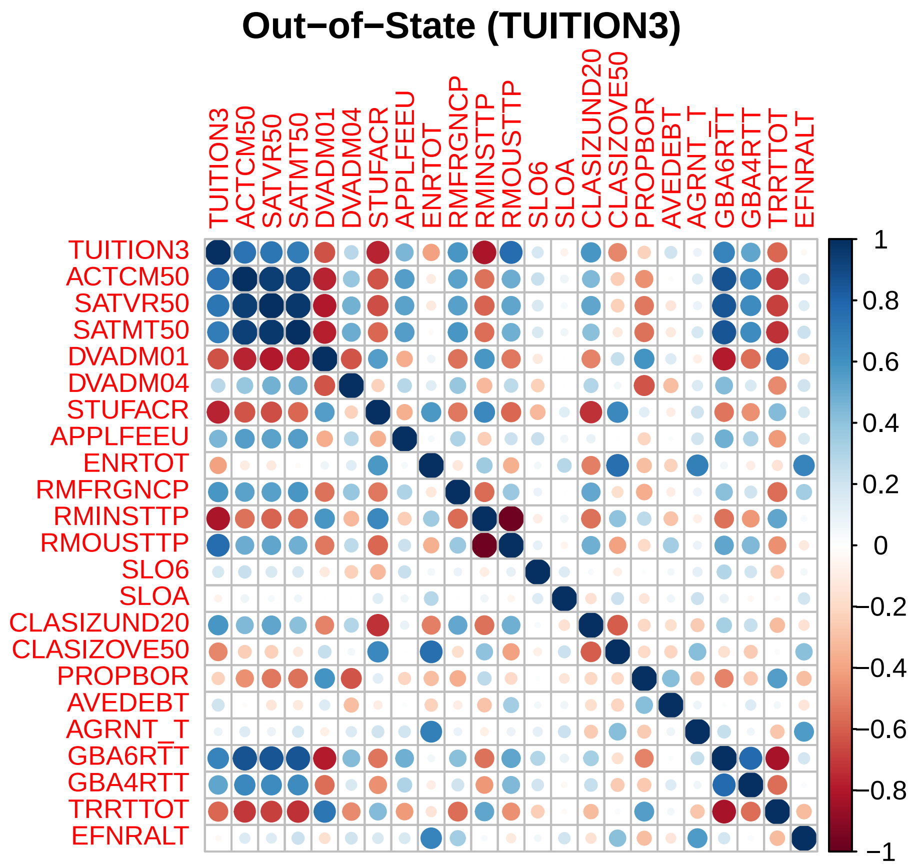
```
VARS_TO_KEEP <- c(
  "CONTROL", "LOCALE", "ACTCM50", "SATVR50", "SATMT50",
  "DVADM01", "DVADM04", "STUFACR", "APPLFEEU", "ENRTOT",
  "RMFRGNCP", "RMINSTTP", "RMOUSTTP", "SL06", "SLOA",
  "CLASIZUND20", "CLASIZOVE50", "PROPBOR", "AVEDEBT",
  "AGRNT_T", "GBA6RTT", "GBA4RTT", "TRRTTOT", "EFNRALT"
)

prep_data <- function(data, target) {
  data %>%
    filter(!is.na(.data[[target]]), .data[[target]] > 0) %>%
    select(all_of(target), all_of(VARS_TO_KEEP)) %>%
    mutate(across(any_of(c("SL06", "SLOA")), ~ ifelse(. == "Yes", 1, 0))) %>%
    select(where(is.numeric)) %>%
    na.omit()
}

get_best_subsets_modern <- function(df_clean, target_var) {
  form <- as.formula(paste(target_var, "~ ."))
  reg_fit <- regsubsets(form, data = df_clean, nvmax = 10, nbest = 1)
  res_sum <- summary(reg_fit)

  data.frame(
    NumVars = 1:length(res_sum$adjr2),
    AdjR2   = res_sum$adjr2,
    res_sum$which
  ) %>% arrange(desc(AdjR2))
}
```

Corr Plot

```r
# Generate the correlation matrices
df_in  <- prep_data(df, "TUITION2")
df_out <- prep_data(df, "TUITION3")

corrplot(cor(df_in), title = "In-State (TUITION2)", mar = c(0,0,1,0), tl.cex = 0.8)
```

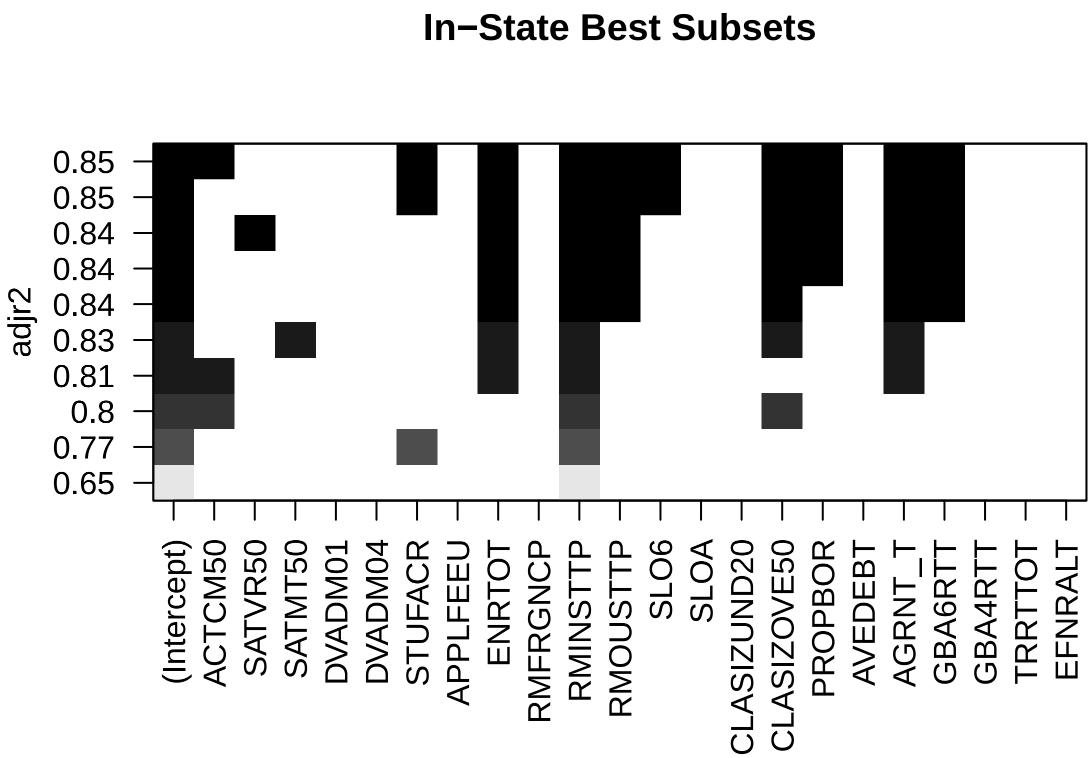



```r
corrplot(cor(df_out), title = "Out-of-State (TUITION3)", mar = c(0,0,1,0), tl.cex = 0.8)
```

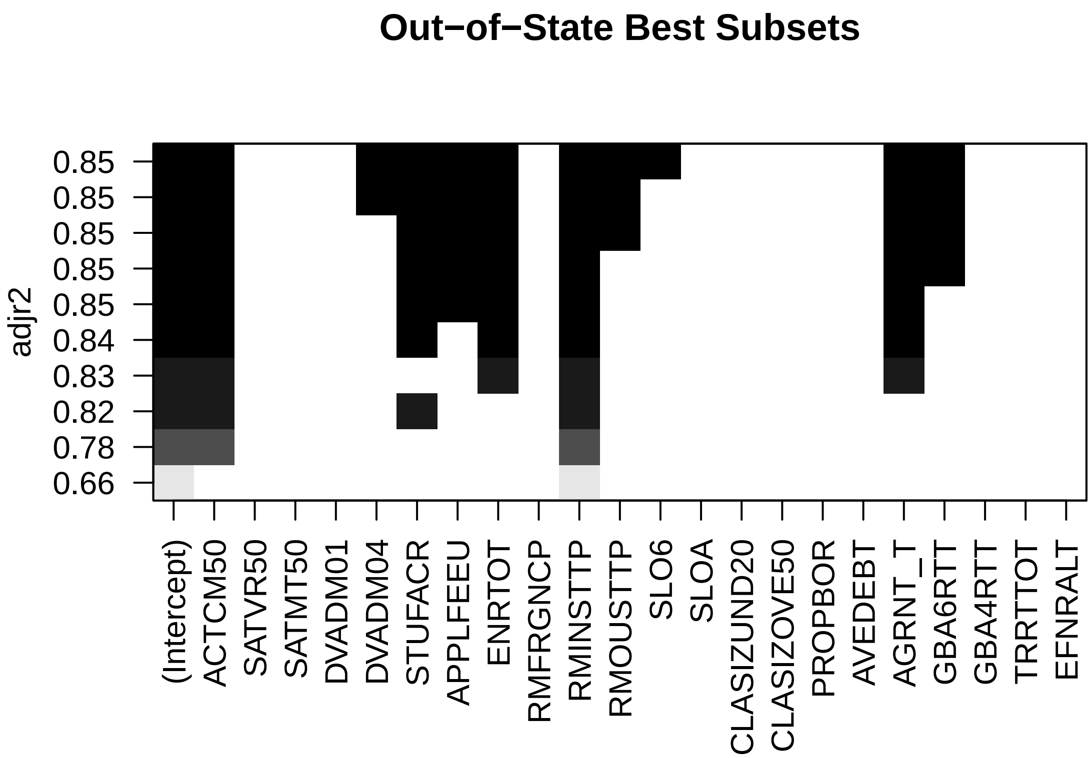


Best Subsets

```
# Run the Analysis
best_in  <- get_best_subsets_modern(df_in, "TUITION2")
best_out <- get_best_subsets_modern(df_out, "TUITION3")

# Visual comparison of models
plot(regsubsets(TUITION2 ~ ., data = df_in, nvmax=10), scale="adjr2", main="In-State Best Subsets")
```




```
plot(regsubsets(TUITION3 ~ ., data = df_out, nvmax=10), scale="adjr2", main="Out-of-State Best Subsets")
```




```r
# Display numerical results
head(best_in, 5)
```

```
##    NumVars     AdjR2 X.Intercept. ACTCM50 SATVR50 SATMT50 DVADM01 DVADM04
## 10      10 0.8508211         TRUE    TRUE   FALSE   FALSE   FALSE   FALSE
## 9        9 0.8491440         TRUE   FALSE   FALSE   FALSE   FALSE   FALSE
## 8        8 0.8447796         TRUE   FALSE    TRUE   FALSE   FALSE   FALSE
## 7        7 0.8399203         TRUE   FALSE   FALSE   FALSE   FALSE   FALSE
## 6        6 0.8365765         TRUE   FALSE   FALSE   FALSE   FALSE   FALSE
##    STUFACR APPLFEEU ENRTOT RMFRGNCP RMINSTTP RMOUSTTP  SLO6  SLOA CLASIZUND20
## 10    TRUE    FALSE   TRUE    FALSE     TRUE     TRUE  TRUE FALSE       FALSE
## 9     TRUE    FALSE   TRUE    FALSE     TRUE     TRUE  TRUE FALSE       FALSE
## 8    FALSE    FALSE   TRUE    FALSE     TRUE     TRUE FALSE FALSE       FALSE
## 7    FALSE    FALSE   TRUE    FALSE     TRUE     TRUE FALSE FALSE       FALSE
## 6    FALSE    FALSE   TRUE    FALSE     TRUE     TRUE FALSE FALSE       FALSE
##    CLASIZOVE50 PROPBOR AVEDEBT AGRNT_T GBA6RTT GBA4RTT TRRTTOT EFNRALT
## 10        TRUE    TRUE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 9         TRUE    TRUE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 8         TRUE    TRUE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 7         TRUE    TRUE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 6         TRUE   FALSE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
```

```r
head(best_out, 5)
```

```
##    NumVars     AdjR2 X.Intercept. ACTCM50 SATVR50 SATMT50 DVADM01 DVADM04
```

```
## 10      10 0.8541484         TRUE    TRUE   FALSE   FALSE   FALSE    TRUE
## 9        9 0.8527058         TRUE    TRUE   FALSE   FALSE   FALSE    TRUE
## 8        8 0.8509133         TRUE    TRUE   FALSE   FALSE   FALSE   FALSE
## 7        7 0.8492592         TRUE    TRUE   FALSE   FALSE   FALSE   FALSE
## 6        6 0.8481619         TRUE    TRUE   FALSE   FALSE   FALSE   FALSE
##    STUFACR APPLFEEU ENRTOT RMFRGNCP RMINSTTP RMOUSTTP  SLO6  SLOA CLASIZUND20
## 10    TRUE     TRUE   TRUE    FALSE     TRUE     TRUE  TRUE FALSE       FALSE
## 9     TRUE     TRUE   TRUE    FALSE     TRUE     TRUE FALSE FALSE       FALSE
## 8     TRUE     TRUE   TRUE    FALSE     TRUE     TRUE FALSE FALSE       FALSE
## 7     TRUE     TRUE   TRUE    FALSE     TRUE    FALSE FALSE FALSE       FALSE
## 6     TRUE     TRUE   TRUE    FALSE     TRUE    FALSE FALSE FALSE       FALSE
##    CLASIZOVE50 PROPBOR AVEDEBT AGRNT_T GBA6RTT GBA4RTT TRRTTOT EFNRALT
## 10       FALSE   FALSE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 9        FALSE   FALSE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 8        FALSE   FALSE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 7        FALSE   FALSE   FALSE    TRUE    TRUE   FALSE   FALSE   FALSE
## 6        FALSE   FALSE   FALSE    TRUE   FALSE   FALSE   FALSE   FALSE
```

Interpretation of Results

The 6, 7, 8, 9, and 10 variables-models all have high Adjusted R^2. The top model is:

```r
summary(lm(TUITION2 ~ ACTCM50 + STUFACR + ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + CLASIZOVE50 + PROPBOR +
```

```
##
## Call:
## lm(formula = TUITION2 ~ ACTCM50 + STUFACR + ENRTOT + RMINSTTP +
##     RMOUSTTP + SLO6 + CLASIZOVE50 + PROPBOR + AGRNT_T + GBA6RTT,
##     data = df_in)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20247.0  -7679.3   -338.2   5998.8  26675.8
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.291e+04  2.163e+04   3.833 0.000197 ***
## ACTCM50      8.251e+02  5.259e+02   1.569 0.119119
## STUFACR     -8.158e+02  3.827e+02  -2.132 0.034921 *
## ENRTOT      -4.664e-01  1.232e-01  -3.784 0.000235 ***
## RMINSTTP    -8.361e+02  1.669e+02  -5.009 1.76e-06 ***
## RMOUSTTP    -5.455e+02  1.742e+02  -3.131 0.002154 **
## SLO6        -3.165e+04  1.199e+04  -2.640 0.009325 **
## CLASIZOVE50 -4.955e+04  1.807e+04  -2.742 0.006968 **
## PROPBOR      1.968e+04  7.497e+03   2.625 0.009709 **
## AGRNT_T      2.656e-04  6.088e-05   4.362 2.61e-05 ***
## GBA6RTT      4.000e+02  1.694e+02   2.361 0.019747 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10090 on 129 degrees of freedom
## Multiple R-squared:  0.8616, Adjusted R-squared:  0.8508
## F-statistic: 80.28 on 10 and 129 DF,  p-value: < 2.2e-16
```

```r
summary(lm(TUITION3 ~ ACTCM50 + DVADM04 + STUFACR + APPLFEEU + ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + AGI
```

```
##
## Call:
## lm(formula = TUITION3 ~ ACTCM50 + DVADM04 + STUFACR + APPLFEEU +
##     ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + AGRNT_T + GBA6RTT,
##     data = df_out)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18086.3  -4451.8    -51.7   3900.1  22826.9
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.054e+04  1.652e+04   3.060 0.002692 **
## ACTCM50      7.454e+02  3.752e+02   1.987 0.049062 *
## DVADM04     -1.162e+02  5.423e+01  -2.143 0.034026 *
## STUFACR     -6.455e+02  2.567e+02  -2.515 0.013144 *
## APPLFEEU     1.371e+02  4.807e+01   2.852 0.005067 **
## ENRTOT      -3.275e-01  7.599e-02  -4.309 3.22e-05 ***
## RMINSTTP    -4.451e+02  1.125e+02  -3.957 0.000125 ***
## RMOUSTTP    -2.171e+02  1.180e+02  -1.839 0.068158 .
## SLO6        -1.398e+04  9.250e+03  -1.512 0.133007
## AGRNT_T      1.826e-04  4.181e-05   4.367 2.56e-05 ***
## GBA6RTT      2.764e+02  1.179e+02   2.344 0.020627 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6971 on 129 degrees of freedom
## Multiple R-squared:  0.8646, Adjusted R-squared:  0.8541
## F-statistic:  82.4 on 10 and 129 DF,  p-value: < 2.2e-16
```

Consistent Drivers (Vertical Columns):

- RMINSTTP (Percent of first-time undergraduates - in-state)
- ENRTOT (Total enrollment)
- AGRNT_T (Total amount of federal, state, local or institutional grant aid awarded to full-time first-time undergraduates)

Key Differences:

- In-State: More sensitive to CLASIZOVE50 (Big class size) than Out-of-state
- Out-of-State: More sensitive to ACTCM50 (ACT Composite 50th percentile score) and APPLFEEU (Undergraduate program application fee) than In-state